What Missionary Control Left in the Language of Two 1880s isiXhosa Newspapers

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Abstract

Where did missionary control leave its trace in the earliest isiXhosa press: in what papers covered (salience) or how they used words (framing)? We compare the missionary *Isigidimi sama-Xosa* with *Imvo Zabantsundu*, founded free of missionary control. Salience separates the papers: Imvo uses parliamentary vocabulary 3.3 times as often; Isigidimi religious vocabulary four times as often (exploratory). Neither survives correction for 15 tests; the religious gap exceeds era drift and matches the archive's account, the parliamentary gap exceeds drift on its point estimate only. Framing evidence is weaker: embedding spaces differ globally and no single concept is robustly shifted; a lean of political words towards religious contexts in Isigidimi survives correction at narrow windows but rests on one word. Historical text costs modern tokenizers 2–12% more subwords per character; spelling alone raises it only for the African-language model. The papers differ in what they printed; whether missionary control changed how political language was used is below our test's detection floor.

Introduction and Background

Where did missionary control leave its trace in the language of the earliest isiXhosa press? The IsiXhosa Intellectual Traditions archive (Schoots et al., 2023a) holds South Africa's earliest African-language newspapers. *Isigidimi sama-Xosa* was a missionary paper; *Imvo Zabantsundu*, founded in 1884 by Isigidimi's former editor J. T. Jabavu (Schoots, 2023b, p. 71), was the first African-owned paper, free "from any missionary control" (Schoots et al., 2023a, p. 55). The archive's authors see Isigidimi's origin "revealed in the frequency of church and Bible stories" (Schoots et al., 2023a, p. 57). We add a controlled test, using the communication-research distinction between *agenda-setting*, what is covered (McCombs & Shaw, 1972), and *framing*, how it is presented (Entman, 1993). We measure the first as **salience** (a concept's vocabulary rate), the second as the company a word keeps in embedding space (Hamilton et al., 2016a). Computational work separating them is mainly on modern, high-resource news (Card et al., 2015; Field et al., 2018). For these papers, Schoots (2023b) tracked root-word frequencies, showing how editors and elections shaped identity vocabulary. We add frequency-matched baselines for salience, an embedding-based framing test against permuted and matched nulls, and an audit of what historical spelling costs modern tokenizers, which already charge low-resource languages more (Petrov et al., 2023; Rust et al., 2021). Who builds such technology, and on whose terms, is itself contested (Bird, 2020). **H1** Salience of political and religious vocabulary differs. **H2** Framing of political concepts differs, Isigidimi's lying closer to religious contexts. **H3** Historical text and its spelling raise tokenizer cost.

Methods

Data. We use the isiXhosa blocks of the archive's human-corrected transcriptions (not spell-checked; orthography and word division are unstandardised; Schoots et al., 2023a, p. 59). The window runs from Imvo's first issue (1 November 1884) to 1888; Isigidimi's 1886 editions are missing (Schoots et al., 2023a, p. 59). Gqoba edited Isigidimi and Jabavu Imvo in this window (Schoots, 2023b, p. 71), so both had African editors; ownership differs. Isigidimi has 286,663 tokens, Imvo 1.31M; per month we draw Imvo pages matching Isigidimi's size, ten times.

Salience. Concepts are stems (Table S1) chosen and checked as in Schoots (2023b), inspecting the most frequent matched forms for false positives: 10 political concepts fixed before testing, four neutral controls, and exploratory religion (*tixo*, *kristu*): anticipated by the archive (Schoots et al., 2023a, p. 57) but selected by an issue-grouped nested-CV classifier trained on the test window (0.83 balanced accuracy). Rate ratios carry issue-bootstrap CIs and are ranked against 2000 matched sets (random word types of the same combined frequency; cf. Monroe et al., 2008); the 15 tests are uncorrected. Held-out windows: Isigidimi 1876–83, Imvo 1889–94.

Framing. We use fastText skip-gram (Bojanowski et al., 2017), more stable than word2vec here (0.83 vs 0.64). Hyperparameters (dim 400, window 1, 20 epochs) were tuned for seed stability (Antoniak & Mimno, 2018) on out-of-window text. A word's *profile shift* is $1 - r$ between its cosine profiles over the 500 most frequent shared words in the two spaces (adapted from Hamilton et al., 2016b); a concept's shift is the mean of its forms' shifts. The null pools issues from both papers and reassigns them to two token-balanced groups (50 permutations; cf. Dubossarsky et al., 2017); concepts are also ranked against 2000 matched sets. Neighbour overlap (Jaccard) follows Gonen et al. (2020); a directional test compares similarity to religious versus political anchors.

Tokenizers. On modern isiXhosa news (Adelani et al., 2023) we count subwords for XLM-R (Conneau et al., 2020), Serengeti (Adebara et al., 2023) and mBERT for (a) modern text, (b) the same with aspiration stripped and (c) historical text; (a to b) per word, (a to c) per character, since historical word division differs.

Results

Salience (H1). Imvo uses parliamentary vocabulary 3.3$\times$ as often ($p = 0.018$) and Isigidimi religious vocabulary four times as often (0.24$\times$, $p = 0.020$); neither survives correction for 15 tests ($p < 0.003$) and nothing else lies beyond the matched



Figure 1: Agenda (left) is a clearer signal than framing (right). Left: lmvo/Isigidimi rate ratio (95% CI); filled = $p < 0.05$ uncorrected. Right (markers as left): profile-shift percentile among matched sets; 'too rare' = too few occurrences, 'seed word' = defines the religious anchors, so untestable.

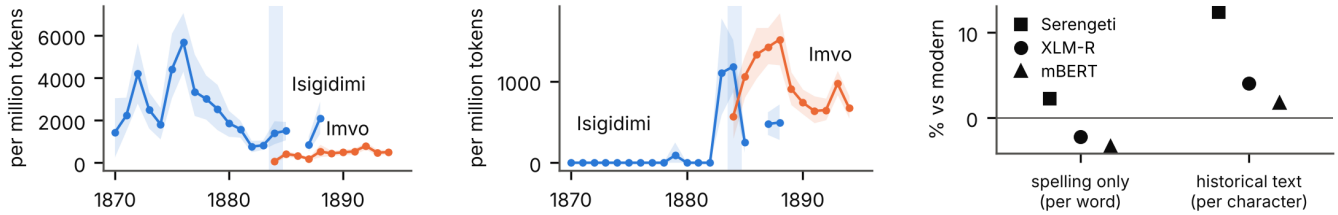


Figure 2: (a) Religious vocabulary (per million tokens) was falling in Isigidimi long before 1884. (b) Parliamentary vocabulary (per million tokens) appears in 1883; light blue: fortnightly issues (Schoots et al., 2023a, p. 59). (c) Subword cost vs modern text: spelling alone (per word) raises it only for Serengeti; historical text (per character) for all.

sets (Fig. 1, left). Isigidimi's parliamentary drift is 1.8× (CI to 5.0, overlapping the gap); *vote* rises 8.5× within Isigidimi, so it and government are not interpreted (Table S3).

Framing (H2). Mean profile shift (0.348) exceeds the permutation null (0.230), a global gap of 0.118, so concepts face matched sets (Fig. 1, right). No single concept is robust: parliament reaches the 96th percentile ($p = 0.045$), as a neutral control (cattle) does. Pooled parliament and vote shift on neighbour overlap (Jaccard) in all five settings (94th–98th percentile, $p = 0.016$–0.061) but on profile only at windows 1–2 (62nd at window 5; Table S2). Political concepts lean towards religious anchors in Isigidimi in all five settings (pooled Δ +0.44 to +0.76; $p < 0.001$ at windows 1–2, surviving Bonferroni over five settings, $p < 0.01$; $p \leq 0.016$ elsewhere, which does not), but without *land* p is 0.010–0.256 and *house* shifts as much. A one-word concept must shift by 0.15 (1.645 matched-set SDs), above the global gap.

Tokenizers (H3). Per word, stripping aspiration raises Serengeti's cost by 2.2%, lowers XLM-R's and mBERT's (-2.2%, -3.4%); per character, historical text costs 12.4%, 4.1% and 1.7% more for Serengeti, XLM-R and mBERT (CIs $\pm 0.5\%$; Fig. 2c).

Discussion

Agenda more than framing (H1, H2). The papers differ clearly in what they printed, as missionary control would predict; whether it also changed how contributors wrote, we cannot show. The religious gap matches the archive's account (Schoots et al., 2023a, p. 57); the parliamentary gap has no such corroboration. Neither is causal: Isigidimi's religious vocabulary fell from 1876, long before Imvo, and rebounded in 1884–85, coinciding with Gqoba's editorship and Imvo's competition; parliamentary vocabulary enters Isigidimi in 1883–84, late in Jabavu's 1881–84 editorship (Schoots, 2023b, p. 71), when it appeared fortnightly (Schoots et al., 2023a, p. 59) and before the 1884 election; editor and election effects cannot be separated. A gap only in editorials would show the editors' voice; one in letters too, different contributor bases (editors' selection or writers' choice, inseparable in print).

What the embeddings see (H2). Words shift globally, not specifically; joining partly disjunctive forms (*uku ba*) left the gap unchanged (0.118 in both). Narrow windows favour grammatical similarity; local context may differ while overall position barely does. The lean rests on *land* (*umhlaba*, also 'earth', 'world'): for a speaker to check.

Responsible NLP (H3). Spelling alone costs only Serengeti, whose vocabulary has twice XLM-R's share of *th*/*kh*/*ph* entries (1.9% vs 0.9%): a one-model hypothesis, inflated if MasakhaNEWS was in its training data. Normalising spelling would erase the orthography the archive preserves and is irreversible (old *t* may be modern *t* or *th*). Shifts describe these editor-filtered papers, not isiXhosa speakers, and cultural readings without speaker review would be extractive (Bird, 2020). The archive's planned machine-learning labelling (Schoots et al., 2023a) risks the same. Model-based framing analyses of this archive would inherit this cost. Limitations: two papers, one window, a missing year, no editor metadata, era-confounded held-out windows, a modern-news domain mismatch, unreviewed stems and glosses (*rulumente*: 'government' here, 'parliament' in Schoots et al., 2023a, p. 62). Whether missionary control changed political language lies below our test's detection floor.

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Statement on the use of generative AI (Faculty of Science AI Ethics Statement)

Generative AI (Claude, by Anthropic, used through the Claude Code tool) helped to write the analysis code (data preparation, embedding experiments, statistics, figure scripts), to draft and edit the text and LaTeX source, and to fill in the checklist below from the course template. Generative AI was also used to review drafts and suggest revisions; suggested changes were checked against the code outputs and the cited sources before being kept. All numbers in the paper are produced by the submitted code and were read from its output files. The authors take full responsibility for all code and text submitted, including the accuracy of every claim and citation.

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    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: New assets are the analysis code and derived word lists (anchors, word-joining rules, the speaker-check sheet), documented in README.txt; no new text corpus is released.

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Supplementary material

S1. Concepts, stems and the word forms they match (1 Nov 1884–1888, both papers)

Stems are matched as substrings of surface forms; following Schoots (2023b) we inspected forms for false positives (our first *law* stem also matched *inteto*, speech, and was dropped) but did not hand-filter them, so run-together forms (e.g. *rulumenteningomsebenzi*) and missed variants remain. All stems, forms and glosses await review by a first-language isiXhosa speaker. Political concepts were fixed before testing; religion was suggested by the classifier and is exploratory.

Concept	Type	Stems	Most frequent matched forms (count; number of distinct forms)
parliament	political	palamente	palamente (1754), ipalamente (47), palamenteni (22), epalamente (14), bapalamente (1), nepalamente (1), yepalamente (1); 11 forms
vote	political	voti	ivoti (301), voti (217), evotini (136), abavoti (98), ngevoti (55), levoti (22), nevoti (22); 115 forms
government	political	rulumente	rulumente (1882), rulumenteni (7), urulumente (5), serulumente (1), rulumenteningomsebenzi (1), zikarulumente (1), rulumenteayiko (1); 8 forms
rule	political	mbuso	zombuso (169), umbuso (76), mbuso (60), yombuso (42), kombuso (37), lombuso (28), kulombuso (23); 43 forms
law	political	mteto	umteto (724), lomteto (475), mteto (406), ngomteto (167), ngumteto (94), komteto (76), nomteto (76); 82 forms
rights	political	lungelo	amalungelo (315), ilungelo (288), lungelo (113), nelungelo (54), namalungelo (50), lilungelo (39), elilungelo (37); 115 forms
country	political	lizwe	lizwe (646), ilizwe (292), kwelilizwe (201), elilizwe (101), elizwe (96), belilizwe (76), yelilizwe (52); 78 forms
land	political	mhlaba	umhlaba (836), mhlaba (208), lomhlaba (156), nomhlaba (105), komhlaba (99), somhlaba (95), yomhlaba (94); 135 forms
freedom	political	kululek	inkululeko (27), kukululekile (24), ukululekile (13), ngenkululeko (10), abakulekileyo (8), nkululeko (8), nenkululeko (7); 92 forms
humanity	political	buntu	ubuntu (45), yobuntu (14), nobuntu (8), sobuntu (8), buntu (7), lobuntu (7), zobuntu (6); 29 forms
time	neutral	xesha	ixesha (787), ngexesha (364), xesha (298), amaxesha (243), lixesha (191), ngamaxesha (149), zamaxesha (114); 207 forms
house	neutral	ndlu	indlu (719), izindlu (328), kwindlu (276), ndlu (234), yendlu (227), ngendlu (133), nendlu (130); 193 forms
rain	neutral	mvula	imvula (267), mvula (31), yemvula (23), ezimvula (17), lemvula (16), yimvula (14), zimvula (10); 45 forms
cattle	neutral	nkomo	inkomo (555), zenkomo (258), nkomo (193), yenkomo (93), nenkomo (75), ngenkomo (35), lenkomo (33); 79 forms
religion	exploratory	tixo, kristu	tixo (684), kristu (207), makristu (7), kristukazi (7), mkristu (3), amakristu (2), lobukristu (1); 16 forms

S2. Secondary framing tests across settings

Percentile of the observed shift among 2000 frequency-matched word sets, and one-sided p (uncorrected). P+V = *palamente* + *voti*; P+V+R adds *rulumente* (uncertain meaning). Directional Δ: Isigidimi minus Imvo, religious minus political similarity, political concepts, two-sided p against matched sets.

Setting	P+V profile	P+V Jaccard	P+V+R profile	P+V+R Jaccard	Δ	Δ p	p w/o land
window 1 (tuned)	92 (p=0.083)	96 (p=0.039)	92 (p=0.077)	99 (p=0.007)	+0.68	$<$0.001	0.010
window 1, forms joined	99 (p=0.010)	95 (p=0.053)	100 (p=0.001)	100 (p=0.002)	+0.44	0.016	0.088
window 2	86 (p=0.139)	94 (p=0.061)	90 (p=0.096)	99 (p=0.010)	+0.76	$<$0.001	0.045
window 2, forms joined	91 (p=0.094)	98 (p=0.023)	99 (p=0.011)	100 (p=0.002)	+0.58	0.009	0.064
window 5	62 (p=0.382)	98 (p=0.016)	81 (p=0.189)	100 (p=0.002)	+0.66	0.015	0.256

S3. Era drift: does a concept's rate move across eras within one paper?

Per-million-token rates and ratios (later/earlier, or Imvo/Isigidimi) with issue-bootstrap 95% CIs.

Concept	Comparison	First	Second	Ratio (95% CI)
parliament	Isigidimi: later / earlier era	214	394	1.84 (0.96–4.98)
parliament	Imvo: later / earlier era	1320	758	0.57 (0.50–0.66)
parliament	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	214	758	3.54 (1.98–9.58)
government	Isigidimi: later / earlier era	1667	1088	0.65 (0.50–0.85)
government	Imvo: later / earlier era	1210	452	0.37 (0.31–0.45)
government	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	1667	452	0.27 (0.22–0.34)
vote	Isigidimi: later / earlier era	72	614	8.55 (3.51–18.94)
vote	Imvo: later / earlier era	799	921	1.15 (0.88–1.51)
vote	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	72	921	12.82 (7.85–25.92)
religion	Isigidimi: later / earlier era	2466	1556	0.63 (0.46–0.84)
religion	Imvo: later / earlier era	362	547	1.51 (1.15–2.03)
religion	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2466	547	0.22 (0.18–0.28)

S4. Anchors for the directional test

Anchors are the 15 top-PMI neighbours of each seed pair in held-out text (Isigidimi 1876-83 + Imvo 1889-94), frozen before testing. Religious (15 words; seeds *tixo*, *kristu*): *sikelela*, *sindisa*, *ukumkanikazi*, *inkosazana*, *buka*, *oyingcwele*, *lilizwi*, *msindisi*, *nelizwi*, *kwelizwi*, *wanga*, *lika*, *yesu*, *izono*, *yehova*. Political (15 words; seeds *palamente*, *voti*): *emqulwini*, *wonyulo*, *lonyulo*, *elugqatsweni*, *eziqingata*, *lokunyula*, *kwesase*, *lokuvota*, *endlu*, *kwamagama*, *ukumela*, *izayo*, *ngalomteto*, *nawupina*, *kwabase*. Seed words cannot themselves be tested (Fig. 1, right). Political anchors include *endlu* (‘in the house’), which may favour the neutral control *house*, and *ngalomteto*, which contains the *law* stem.